Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Cartreine, James A. Ph.D.		
Project Title:	Designing a Smart Medical System for Psychosocial Support		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Brigham and Women's Hospital/Harvard Medical Center		
PI Address 1:	Program in Behavioral Informatics and eHealth		
PI Address 2:	Department of Psychiatry		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:	NOTE: PI moved to Brigham and Women's Hospital; formerly at Beth Israel Deaconess Medical Center, Boston, MA, per info received December 2011 (Ed.) Name change to Cartreine in summer 2008 (from Carter), per NSBRI information (11/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	05/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Holland, Albert (NASA JSC)		
Grant/Contract No.:	NCC 9-58-NBPF00207		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	INTRODUCTION Although interpersonal conflict can be productive, unmanaged conflict can pose a threat to the success of long-duration missions. This project involves the development of a conflict management training program to teach long-duration crews how to manage intra-crew conflict, crew-ground conflict, and silent conflict, and to mediate between crewmembers who are in conflict. This training is part of a larger network of self-guided interactive multimedia programs to train and assist long-duration flyers in the prevention, assessment, and management of multiple psychosocial problems that can arise on extended missions. The system is intended for use both during training and on orbit. METHODS The project involves three phases. Phase 1: Background Data Collection. Consultation interviews were conducted with 13 veteran long-duration flyers who have lived on the International Space Station, Mir, and Skylab. Interviewees were presented with five hypothetical scenarios, three of which depicted interpersonal conflicts, both within crews and between space and ground, on extended missions. Input was obtained on the best practices for managing such situations, as well as mistakes that a novice might make in handling them. Input was also obtained from clinical experts in interpersonal conflict management and depression treatment. Content of the training program was based on these data. Phase 2: Interactive Multimedia Production. A Virtual Space Station 3-dimensional graphic was created to serve as a portal to multimedia-based training, assessment, and intervention resources. The system is modular and expandable, to accommodate additional content in the future. Original content is being created to teach how to manage conflicts, and presented via interactive video simulations, activities, and didactics. All training and intervention content is guided by an on-camera coach. Phase 3: Formative Evaluation of the system by astronauts at NASA-Johnson Space Center. Formative evaluation of a prototype of the training will be conducted with 10 members of the astronaut corps. They will use portions of the system and provide feedback regarding its usability, acceptability, and perceived value. This data will guide further program development and revisions. EARTH APPLICATIONS Although this study involves developing countermeasures to assist long-duration flyers, it also provides a model that could be applied in many earthbound settings, both in operational environments and in everyday life.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The system, with simple modifications and revisions, could be adapted for use in other isolated operational environments, such as polar research stations, submarines, commercial ships, oilrigs, and underwater research bases. Furthermore, even greater value could be derived by making similar psychosocial support systems available to the public in settings such as primary care practices, public and mental health centers, schools, social services offices, places of worship, military bases, prisons, and eventually at home or in any location, through broadband Internet.
Task Progress:	In the past year, the countermeasure have been used by Wyle Life Sciences staff in stand-up training for astronauts on conflict management. Additionally, a formative evaluation, focuing on usability and acceptability, was conducted at NASA-Johnson Space Center with 10 current members of the astronaut corps. These data are currently being compiled and formatted at JSC's Usability Testing and Analysis Facility in preparation to be sent to the investigators. The data will be used to make improvements to all aspects of the program, going forward. The conflict management training was also used by Dr. Joseph Brady and his colleagues to train subjects in their Mars mission simulation.
Bibliography Type:	Description: (Last Updated: 02/11/2021)